

Ionic conduction in poly(ethylenimine)– and poly(N-methylethylenimine)–lithium salt systems

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Linear polyethylenimine (PEI) and poly(N-methylethylenimine) (PMEI) (DP=2850) form ion-conducting solid solutions with lithium salts. Four such combinations PEI–LiClO₄, PEI–LiO₃SCF₃, PMEI–LiClO₄, and PMEI–LiO₃SCF₃, were studied. When the salt content is low, PEI systems are more conductive than the corresponding PMEI systems. Toward high salt loadings, the ionic conduction is governed by the bulk viscosity, difference due to the matrix polymers and/or salts diminishes.

1. Introduction

Linear poly(ethylenimine) (PEI) and poly(N-methylethylenimine) (PMEI), being direct nitrogen analogs of poly(ethylene oxide) (PEO), can be eligible candidates for the matrices of polymer electrolytes [1]. A desirable matrix should (1) provide cation–polymer bonds strong enough to dissolve sufficient amount of salt, and yet labile enough to enable the salt to dissociate to release free ionic carriers, (2) give completely amorphous complex films with a low glass transition temperature [2].

A preliminary study on PMEI–LiClO₄ by us appeared in 1984 [3]. Chiang and coworkers [4–6] studied PEI–NaI and several PEI–Li salt systems. Ratner et al. [7] reported on PEI–NaO₃SCF₃ systems. Their observations were qualitatively very analogous to those for polyether–salt systems. Unfortunately, the PEI employed by Chiang was of a low molecular weight (DP=ca. 46); the data for the truly “poly”imine–Li salt systems have been lacking.

In the present paper, we prepared high molecular weight PEI and PMEI that have an equal average chain length. Both polymers were complexed with LiClO₄ and LiO₃SCF₃, the glass transition and the ionic conductivity behavior was compared.

2. Experimental

2.1 Materials

Linear PEI, DP=2850, was prepared by acid hydrolysis of poly(2-phenyl-2-oxazoline) [8]. The PEI was recrystallized twice from 1:1 ethanol-water, and thoroughly rinsed with deionized water to remove inorganic contaminants. Part of the PEI was converted [8] into PMEI to make the chain lengths of both polymers equal. The PEI and PMEI were dried before use by evacuation ($<10^{-2}$ Torr) at 60 and 50°C, respectively, for 72 h.

Reagent grade LiClO₄ was used as received. LiO₃SCF₃ was prepared from Li₂CO₃ and CF₃SO₃H and recrystallized from acetone. Both salts were dried immediately before use under vacuum at 130°C for 48 h.

2.2 Polyimine-lithium salt complexes

A 10% solution of ca. 0.5 g of dry PEI (or PMEI) and a calculated amount of a salt in anhydrous methanol was stirred over night under a dry nitrogen atmosphere. The solvent was removed under reduced pressure till the mixture became viscous, it was cast onto a stainless steel bottom electrode for conduc-

tivity measurements, which also served as a mold for the sample disc, 12 mm in diameter and 0.3 mm in thickness. After most of the methanol was gone, the sample was pressed with a top electrode and shaped into a disc to the size of the mold, and evacuated under 0.01 Torr at 130°C for 12 h.

2.3 Measurements

Glass transition temperatures (T_g) of PEI, PMEI and their composites with lithium salts were determined by a Seiko I&E DSC-20 connected to SSC-570 thermal controller.

Ionic conductivities were determined under vacuum ($<10^{-2}$ Torr) by the complex impedance method using a computer-driven Hewlett-Packard 4276A LCZ meter. The cell consisted of a pair of stainless steel electrodes embedded in a stainless steel heating block, which was enclosed in an evacuable glass container.

3. Results and discussion

3.1 Glass transition temperatures

PEI is a highly crystalline polymer, which melts at ca. 55°C [8,9]. A DSC trace of our PEI shows a clear heat capacity change at -23°C. DSC curves for the PEI-Li salt systems indicate that the PEI crystal melting peak persists at $N/Li=15$, and becomes barely detectable at $N/Li=10$, in agreement with a previous result [5]. PMEI, in contrast to PEI, is a completely amorphous polymer, whose T_g value is as low as -88°C.

Incorporation of a lithium salt causes a large increase in the T_g values in both polyimines, as shown in fig. 1. $LiClO_4$ raises the T_g 's more significantly than LiO_3SCF_3 . Despite the 65°C difference between T_g 's of the pristine polyimines, the T_g 's of pairs (1) $LiClO_4$ -loaded PEI, and PMEI, and (2) LiO_3SCF_3 -loaded PEI and PMEI, respectively, approach each other in the higher salt content regions. At $N/Li=4$, the T_g values of the $LiClO_4$ laden polyimines are higher than those of their LiO_3SCF_3 equivalents by ca. 20°C.

The observed T_g versus composition curves for the PEI systems, given in fig. 1, might seem peculiar in

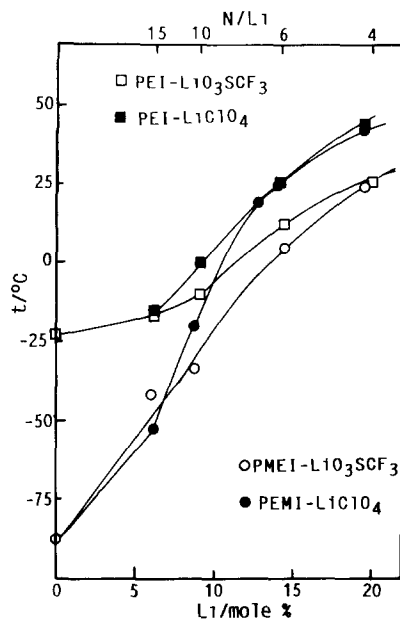


Fig. 1. Plots of glass transition temperature versus composition for PEI- and PMEI-Li salt systems.

the low salt content region, one might suspect the T_g of PEI itself given here to be too high. In fact, by extrapolating a T_g versus composition curve for a series of epoxy-cured, low molecular weight PEI, Chiang et al. [5,6] estimated the T_g of the non-crosslinked PEI to lie near -40°C. One must note, however, that the PEI is highly crystalline by virtue of its extensive hydrogen bonding [9]. Motions of the amorphous regions must also be severely constrained by the hydrogen bonds. Li salts incorporated in PEI serve to raise T_g through effective dipole-ion interactions, but simultaneously destroy the hydrogen bonds. These opposing effects probably balance at $N/Li=15-10$. At higher salt contents, the constraints due to hydrogen bonds disappear, the T_g behavior of PEI systems, thus, becomes to resemble that of PMEI systems.

3.2 Ionic conductivities

The highest conductivity value, $2 \times 10^{-6} \text{ S cm}^{-1}$, at 60°C was obtained with PEI- LiO_3SCF_3 ($N/Li=15$). This is, at best, comparable with only some mediocre values for PEO-Li salt systems [10]. But

the ways the ionic conductivities are controlled by the nature of matrix polymers and/or Li salts are interesting

Fig. 2 compares a pair of PEI systems where the incorporated salts are different. The lower the salt content, the higher the conductivity at all temperatures for both salts. PEI-LiO₃SCF₃ systems are always more conductive than the corresponding PEI-LiClO₄ systems. As the salt content increases, the conductivity difference widens. What fig 2 shows nicely the correspondence with the T_g profiles in fig 1: T_g 's of these couples are comparable when N/Li is 15, and PEI-LiClO₄ has increasingly higher T_g 's as the salt content increases. The conductivity is primarily controlled by the T_g 's, i.e. bulk viscosities, regardless of the counter anion

Fig. 3 offers another comparison when the matrix polymer is PMEI. Here the features are approximately similar to those of fig 2, but the situation is complicated by the existence of maxima around N/Li=6 in the conductivity versus composition plots, as will be seen later in fig 6

Fig 4 compares the effects of matrix polymers while the salt is LiClO₄. At N/Li=4 and 6, where T_g 's of the two systems are practically the same, the

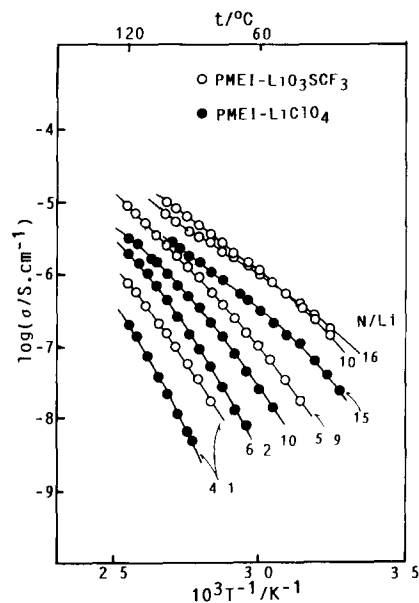


Fig 3 Temperature dependence of ionic conductivities for PMEI-Li salt systems comparison with respect to the anionic species

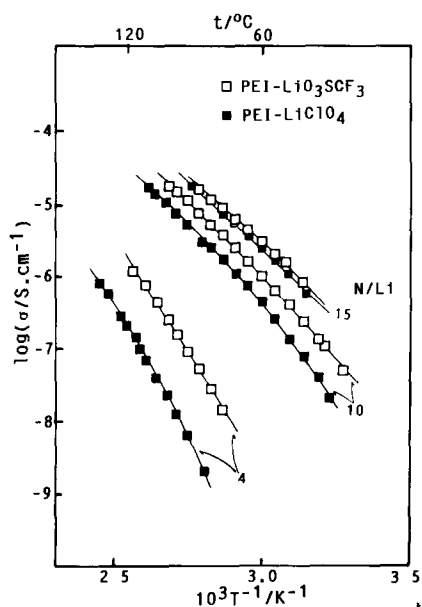


Fig 2 Temperature dependence of ionic conductivities for PEI-Li salt systems comparison with respect to the anionic species

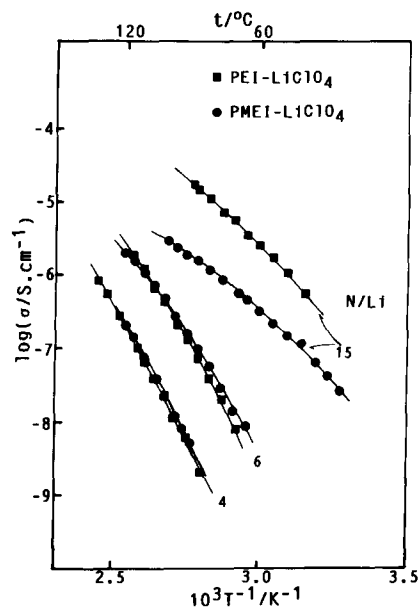


Fig 4 Temperature dependence of ionic conductivities for polymine-LiClO₄ systems comparison with respect to matrix polymers

conductivity curves are almost superimposable. At $N/Li = 15$, however, PEI– $LiClO_4$ is an order of magnitude more conductive than PMEI– $LiClO_4$, although the T_g of the former is higher by $35^\circ C$.

The matrix effect is again seen in fig. 5, while the salt is LiO_3SCF_3 . When $N/Li = ca. 4$, where both systems have similar T_g as fig. 1 shows, their conductivities are almost the same. At $N/Li = 5.9$ and 10 , where the PMEI systems have lower T_g 's, they are more conductive than the PEI system at low temperatures. At higher temperatures, however, the conductivity order is reversed. Finally, at $N/Li = 15–16$, PEI system is definitely more conductive at all temperatures.

The ionic conductivities appear so far to depend primarily on the T_g values of the systems, especially at higher salt contents. The conductivity profile at a reduced temperature, $T - T_g = 80^\circ C$, in fig. 6 immediately shows that the regions of high and low salt contents have to be considered separately.

The conductivity plots in $N/Li = 4–6$ region are located close to each other irrespective of the matrices or salts. The slight decrease in conductivities with the increase in salt content may indicate some

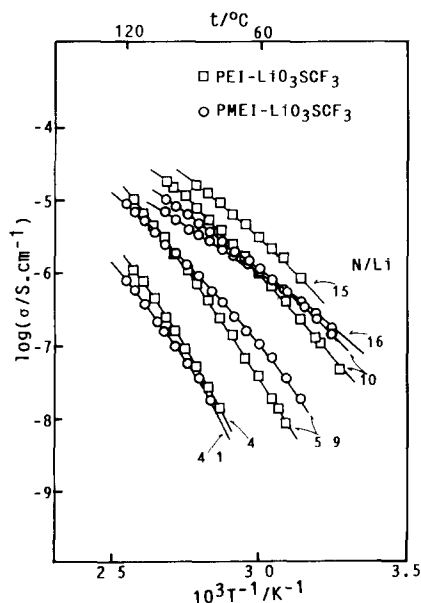


Fig. 5 Temperature dependence of ionic conductivities for polyimide– LiO_3SCF_3 systems: comparison with respect to matrix polymers.

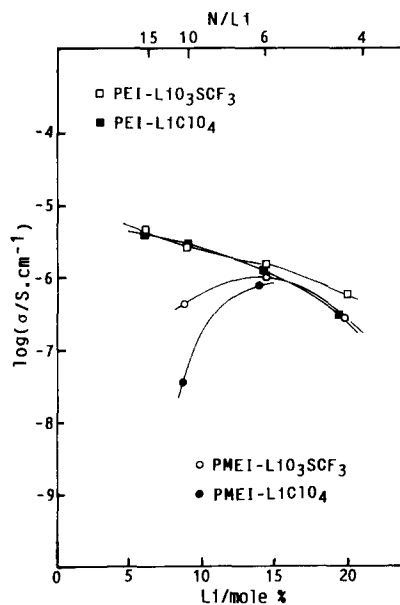


Fig. 6 Plots of ionic conductivities versus compositions for polyimide– Li salt systems at a reduced temperature, $T - T_g = 80^\circ C$.

decrease in the drift mobility of ionic carriers caused by the increased number of ion clusters [11].

In the low salt content region ($N/Li > 10$), PEI and PMEI systems behave quite differently. If the reduced temperature approach has taken most of the mobility factors into account, the difference in conductivities is attributed to the difference in ionic carrier concentrations. The salt dissociating ability of PEI is probably strong enough to create free ionic carriers from both salts, even at a low salt concentration, and at a relatively low temperature. PMEI, probably being a weaker salt dissociating matrix, discriminates between LiO_3SCF_3 and $LiClO_4$; the number of carriers does not reach the “saturation” value until the salt content reaches 15 mol%. Fig. 6, as a whole, suggests that the PMEI systems are very reminiscent of the corresponding amorphous polyether– Li salt systems, whereas the PEI systems differ in the low salt content region. The interplay of the metal ion complexation by the basic nitrogens and the hydrogen bonds due to $N-H$ groups, which constrain the matrix motions and solvate anions as well, is probably responsible.

4. Conclusion

Incorporation of LiClO_4 or LiO_3SCF_3 into PEI and PMEI gives rise to polymer electrolytes similar to PEO–salt systems. In the high salt content region, the ionic conductivities are essentially governed by the bulk viscosity of the systems irrespective of the matrices or salts. In the low salt content region, the PEI systems are much more conductive than the corresponding PMEI systems.

Acknowledgement

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